UNEMPLOYMENT TREND ANALYSIS

EDA MICRO PROJECT

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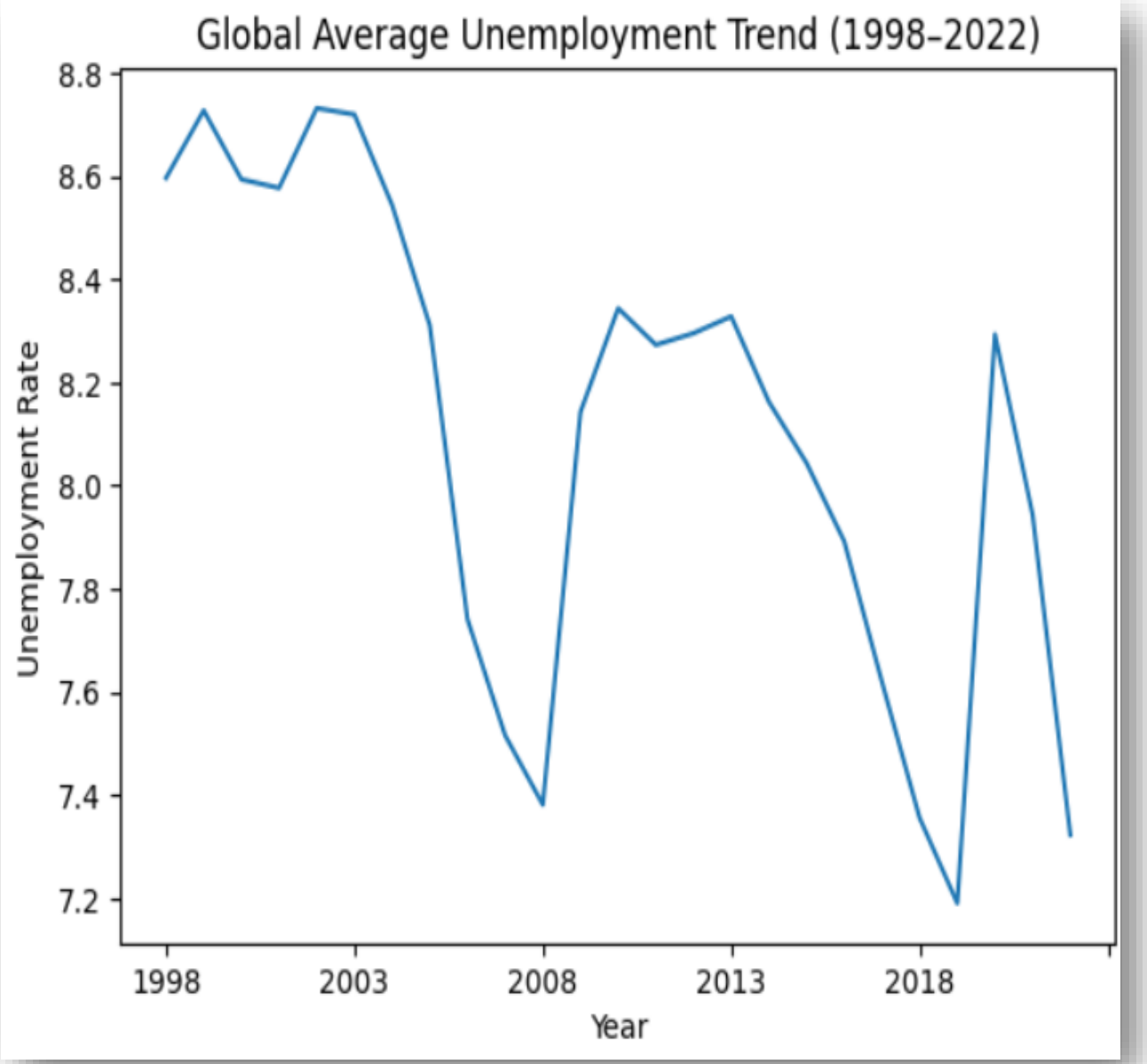
INTRODUCTION

Unemployment shows the percentage of people who are without jobs but actively looking for work. In this project, unemployment data from 1998–2022 is analyzed using Python to understand trends and compare different countries.

TREND ANALYSIS

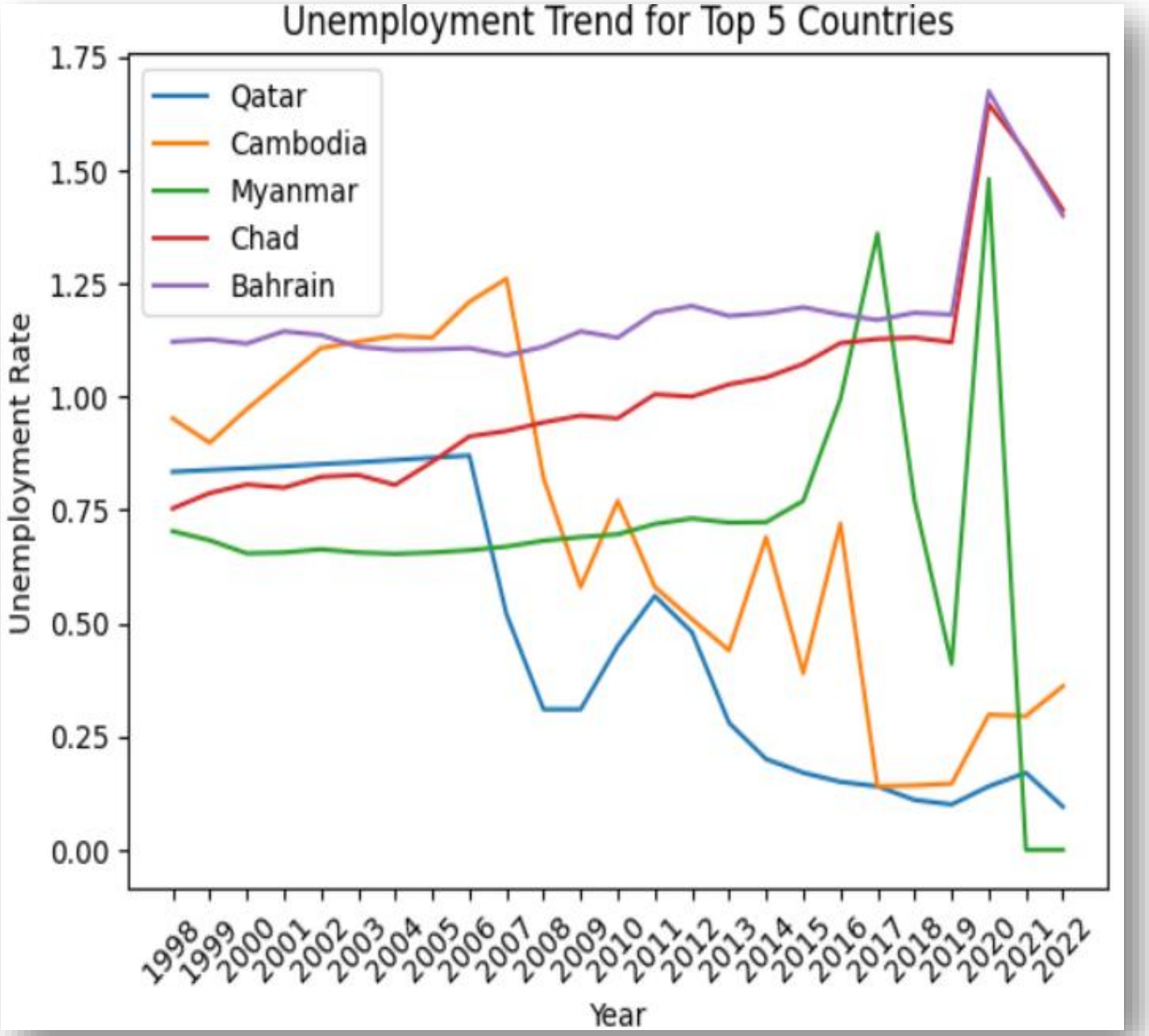
- Global Average Unemployment Trend

- Unemployment was high in the early 2000s (around 8.6–8.7%).
- From 2014 to 2019, unemployment gradually decreased.
- Overall: The unemployment rate changes over time but shows a slight decrease in recent years.



❑ Unemployment Trend for Top 5 Countries

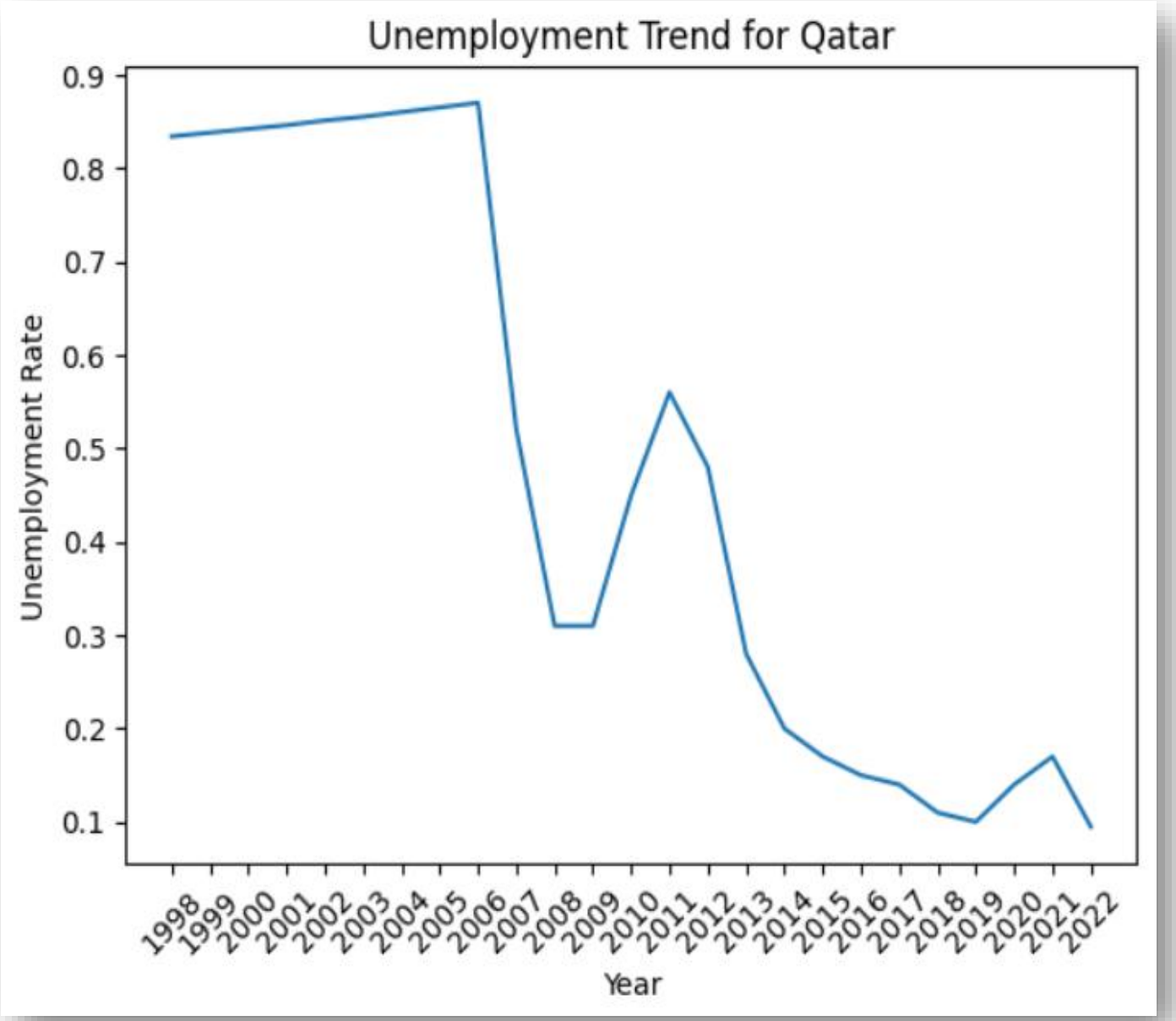
- Qatar: Unemployment decreased over time, reaching very low levels after 2012.
- Cambodia: Rate fluctuated a lot but overall decreased in later years.
- Myanmar: Mostly stable but had sudden spikes around 2017 and 2020.
- Chad: Shows a slow and steady increase in unemployment over the years.
- Bahrain: Fairly stable but increased slightly after 2015.



❑ Unemployment Trend for Qatar

- Qatar's unemployment rate was relatively higher in the early years (around 1998–2005) but started declining after 2006.

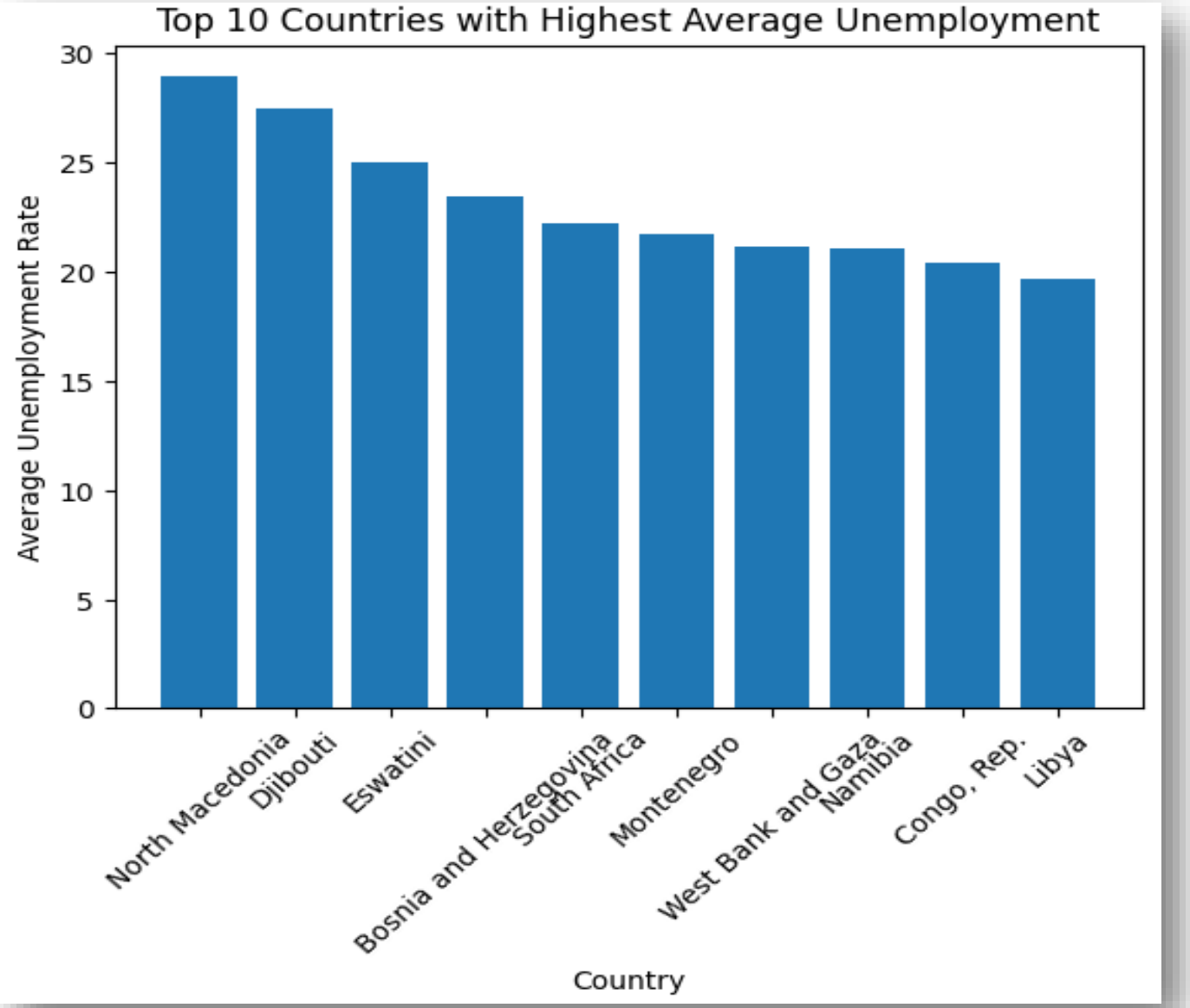
- In recent years, the unemployment rate has remained very low, showing strong employment stability in the country.



COUNTRY COMPARISON

- Top 10 Countries with Highest Average Unemployment

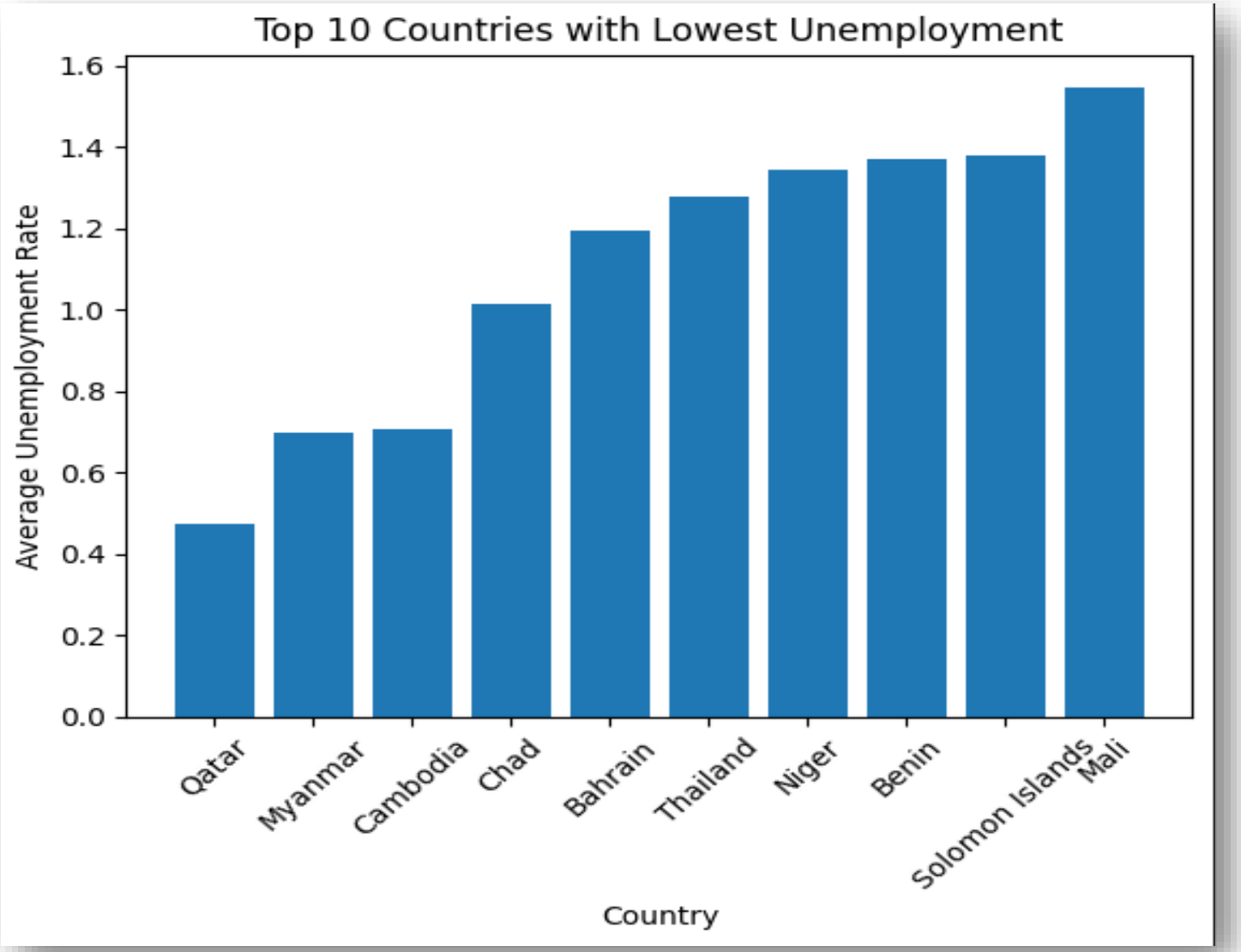
- The graph shows the top 10 countries with the highest average unemployment rates over the years
- North Macedonia and Djibouti have the highest unemployment rates, while the other countries also show relatively high unemployment compared to global levels.



❑ Top 10 Countries with Lowest Unemployment

- The graph shows the top 10 countries with the lowest average unemployment rates. Qatar has the lowest unemployment rate among these countries.

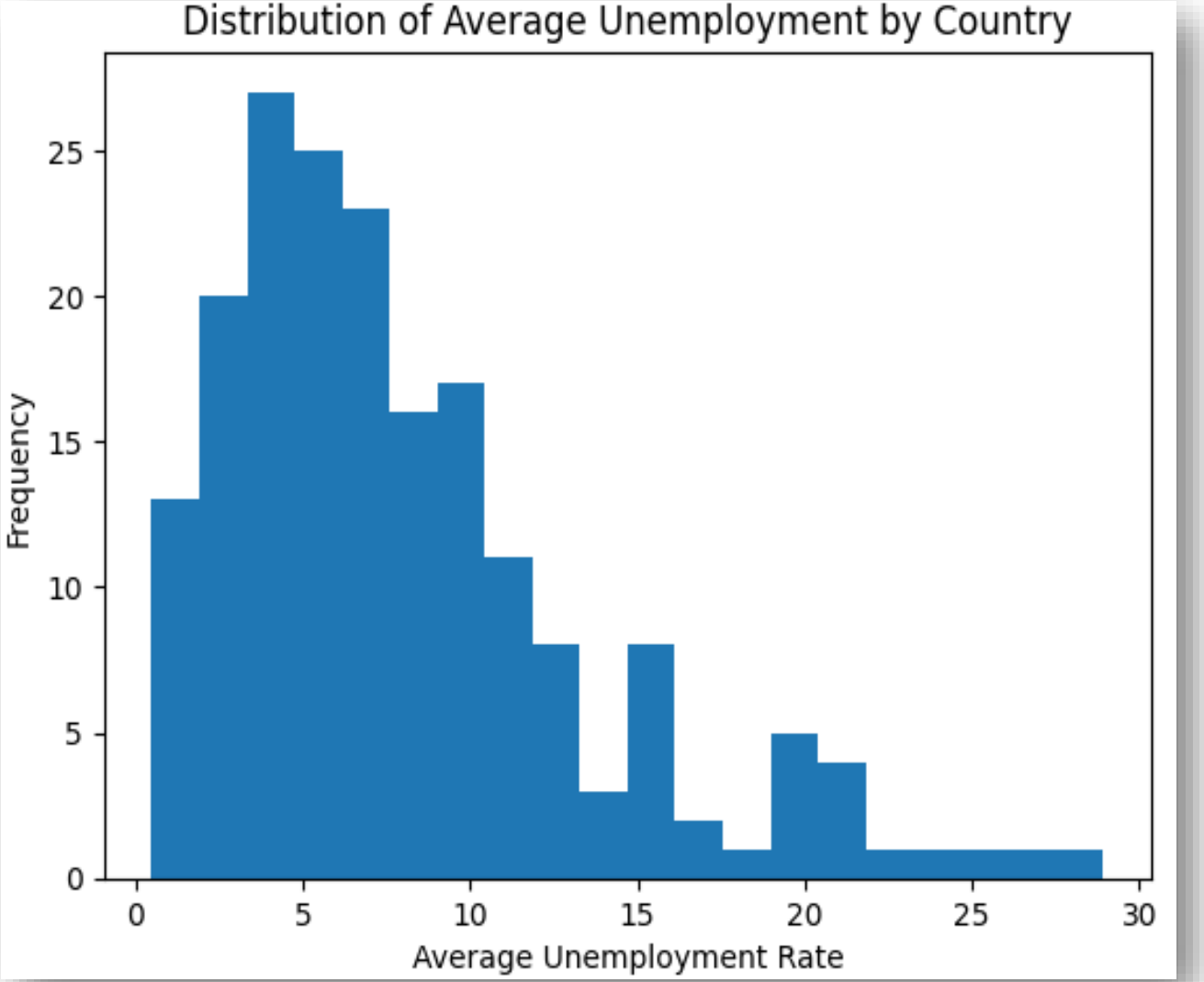
- Other countries such as Myanmar, Cambodia, and Chad also maintain relatively low unemployment levels compared to global averages.



❑ Distribution of Average Unemployment

- The histogram shows the distribution of average unemployment rates across different countries.

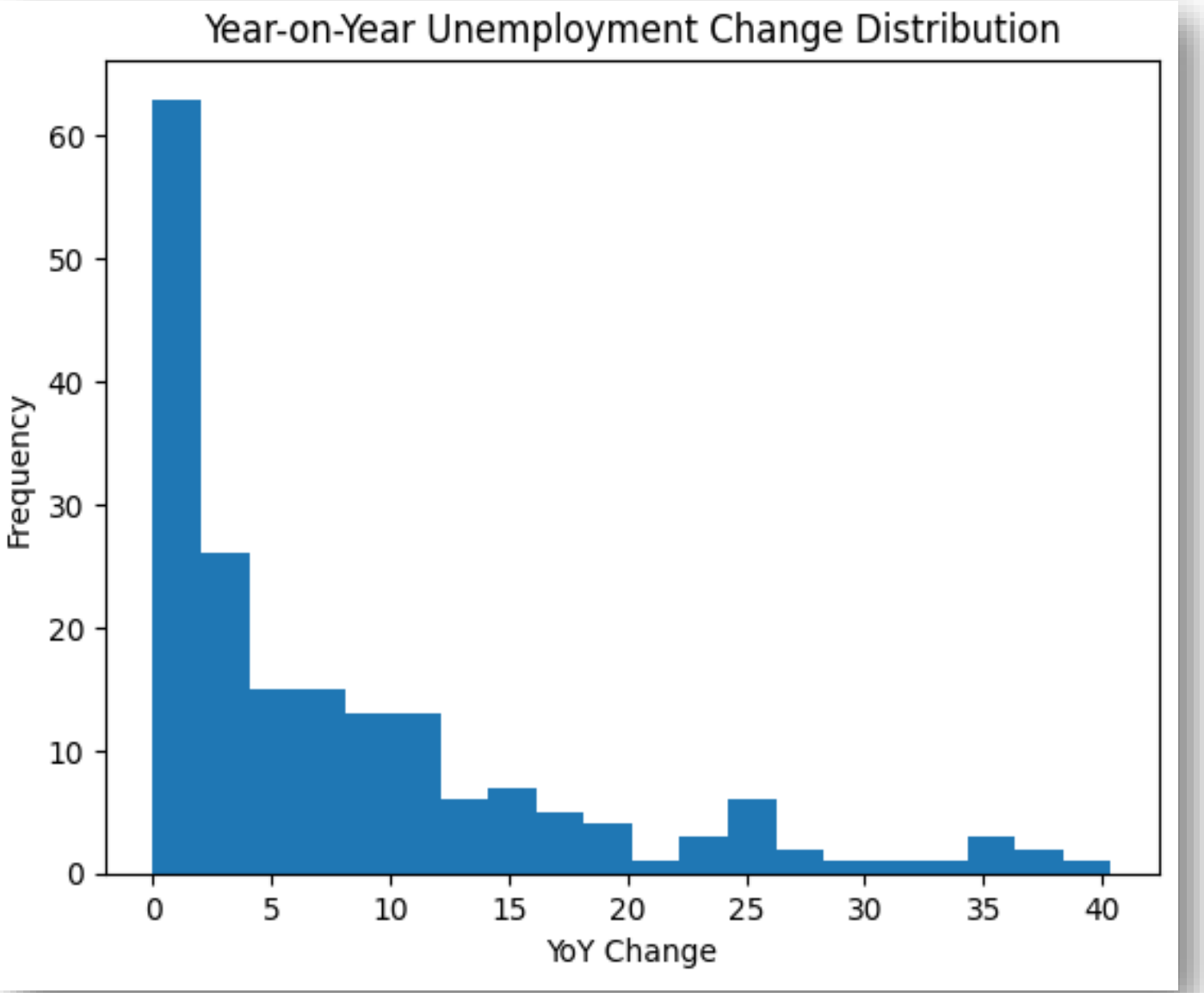
- Most countries have unemployment rates between about 3% and 10%, while only a few countries have very high unemployment rates.



YEAR-ON-YEAR CHANGE ANALYSIS

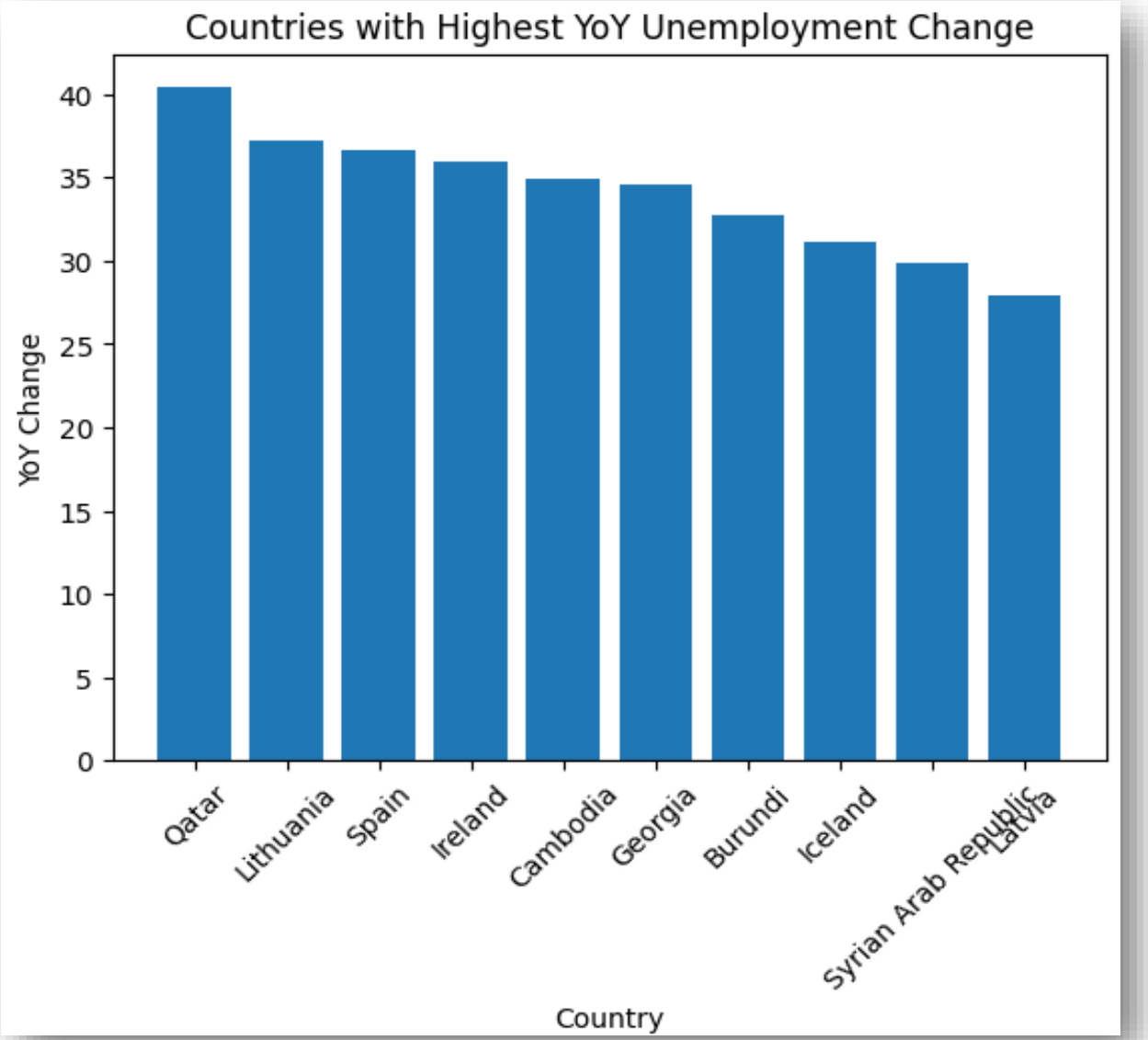
❑ YoY Unemployment Change Distribution

- Most unemployment changes are small (0–10)
- A few years show large increases in unemployment
- This means unemployment is usually stable but sometimes rises sharply during difficult economic periods



❑ Countries with Highest YoY Change

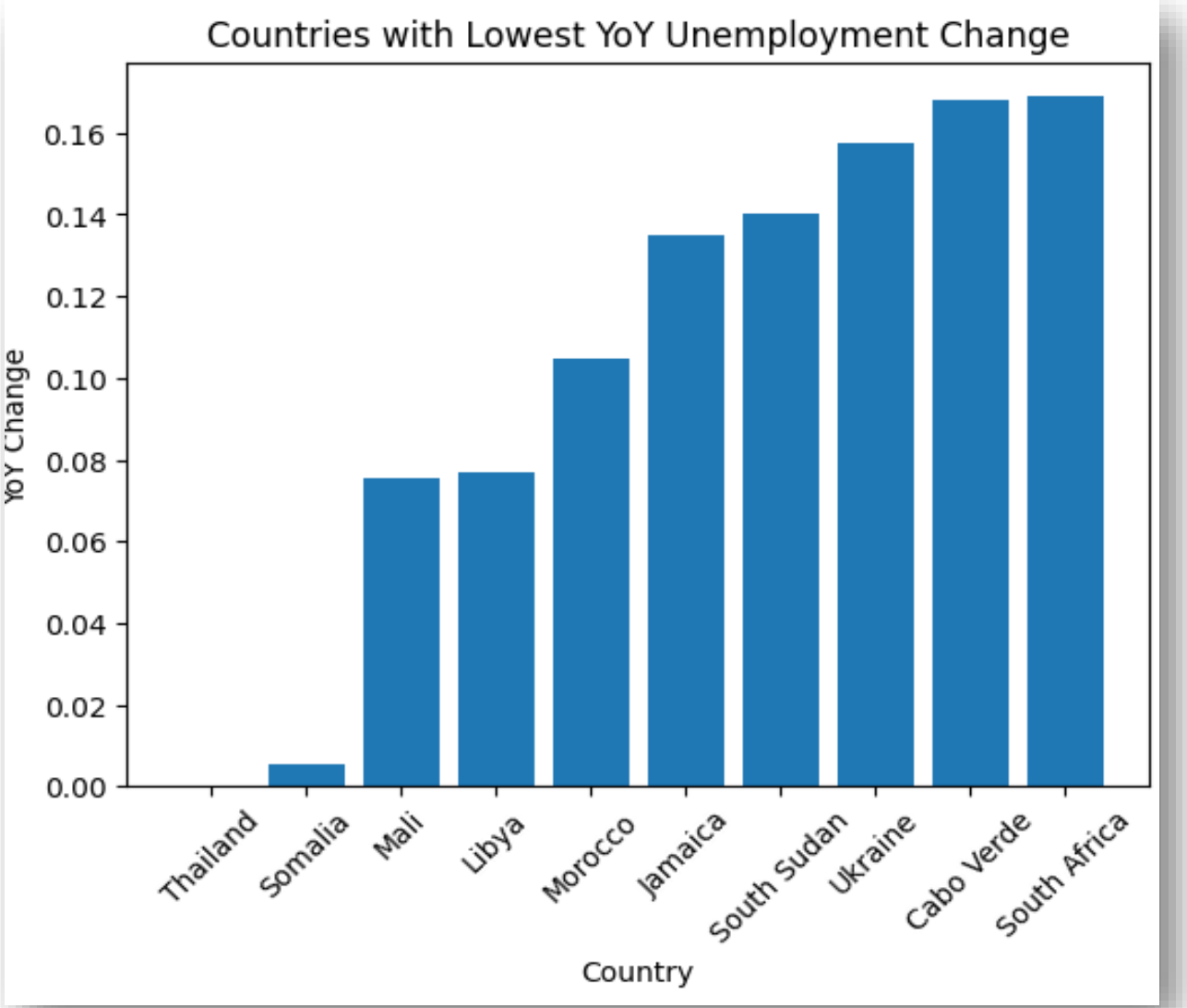
- ➢ Lithuania, Spain, and Ireland also have high changes in unemployment.
- ➢ Some countries experience larger yearly changes in unemployment, showing economic fluctuations.



❑ Countries with Lowest YoY Change

- Thailand has the lowest YoY unemployment change, showing very stable employment.

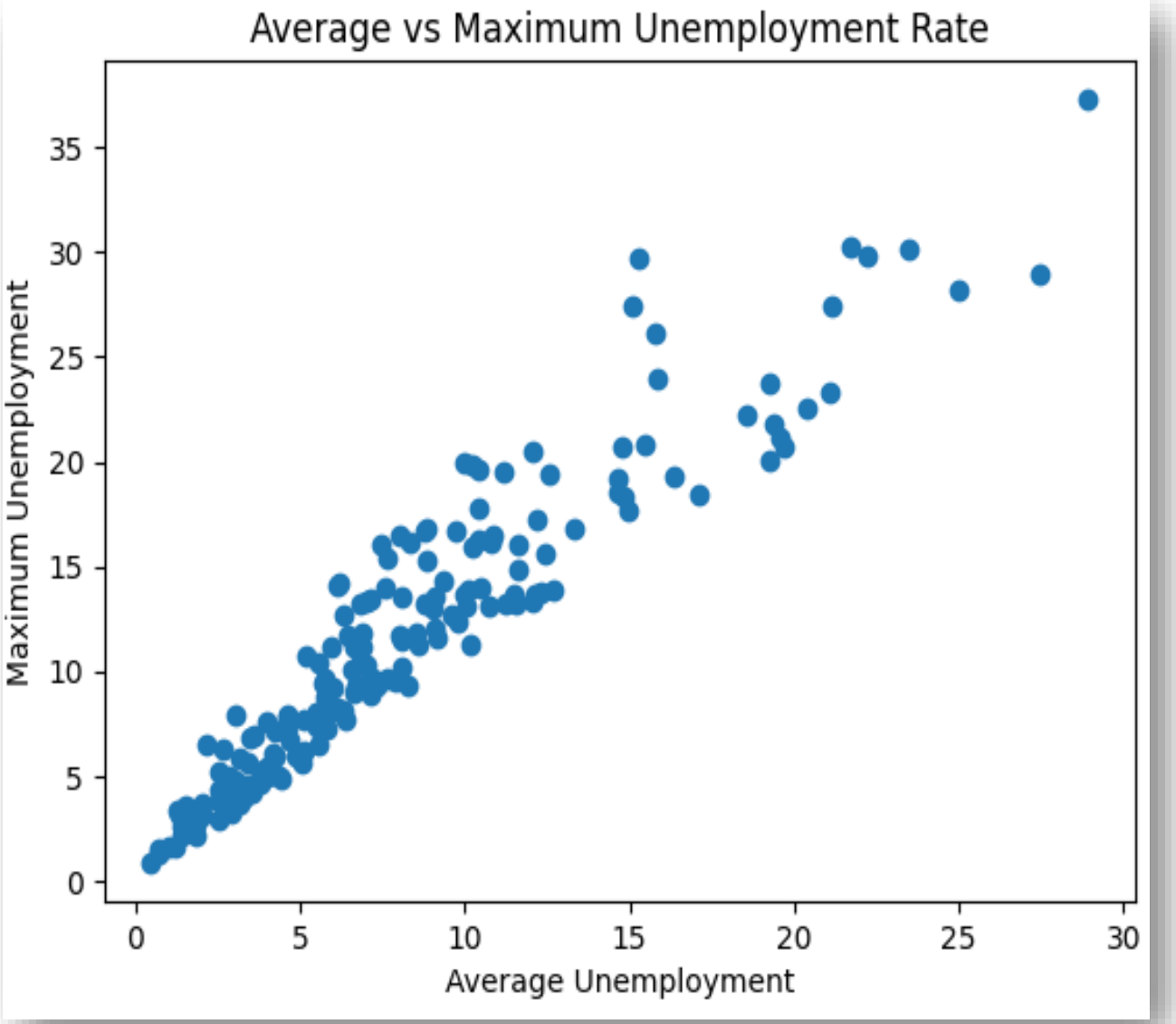
- Overall: These countries have relatively stable unemployment rates with only small yearly changes.



RELATIONSHIP & CORRELATION ANALYSIS

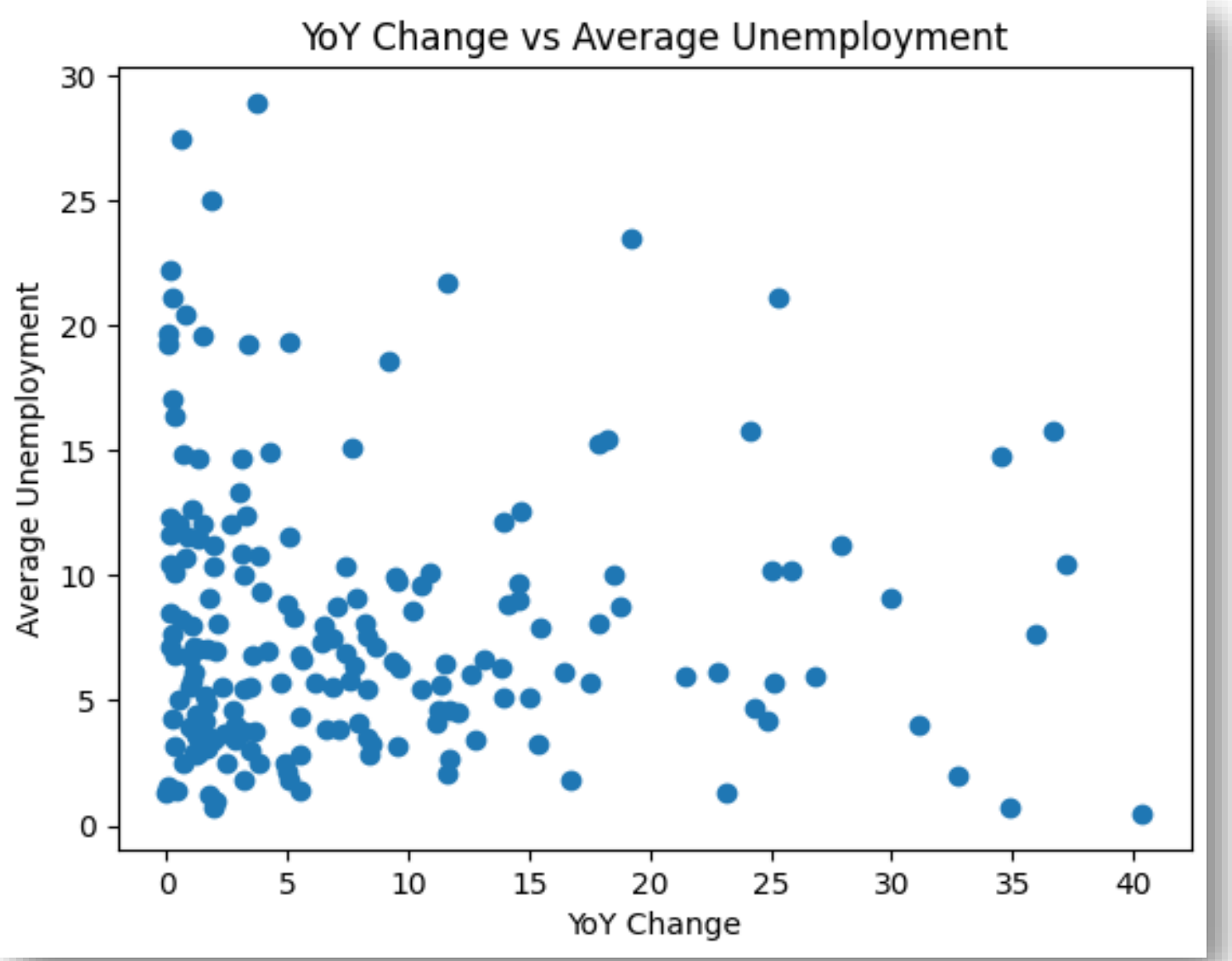
❑ Average vs Maximum Unemployment

- Countries with higher average unemployment usually have higher maximum unemployment.
- Most data points follow an upward trend, showing a positive relationship.
- There is a strong positive correlation between average and maximum unemployment rates.



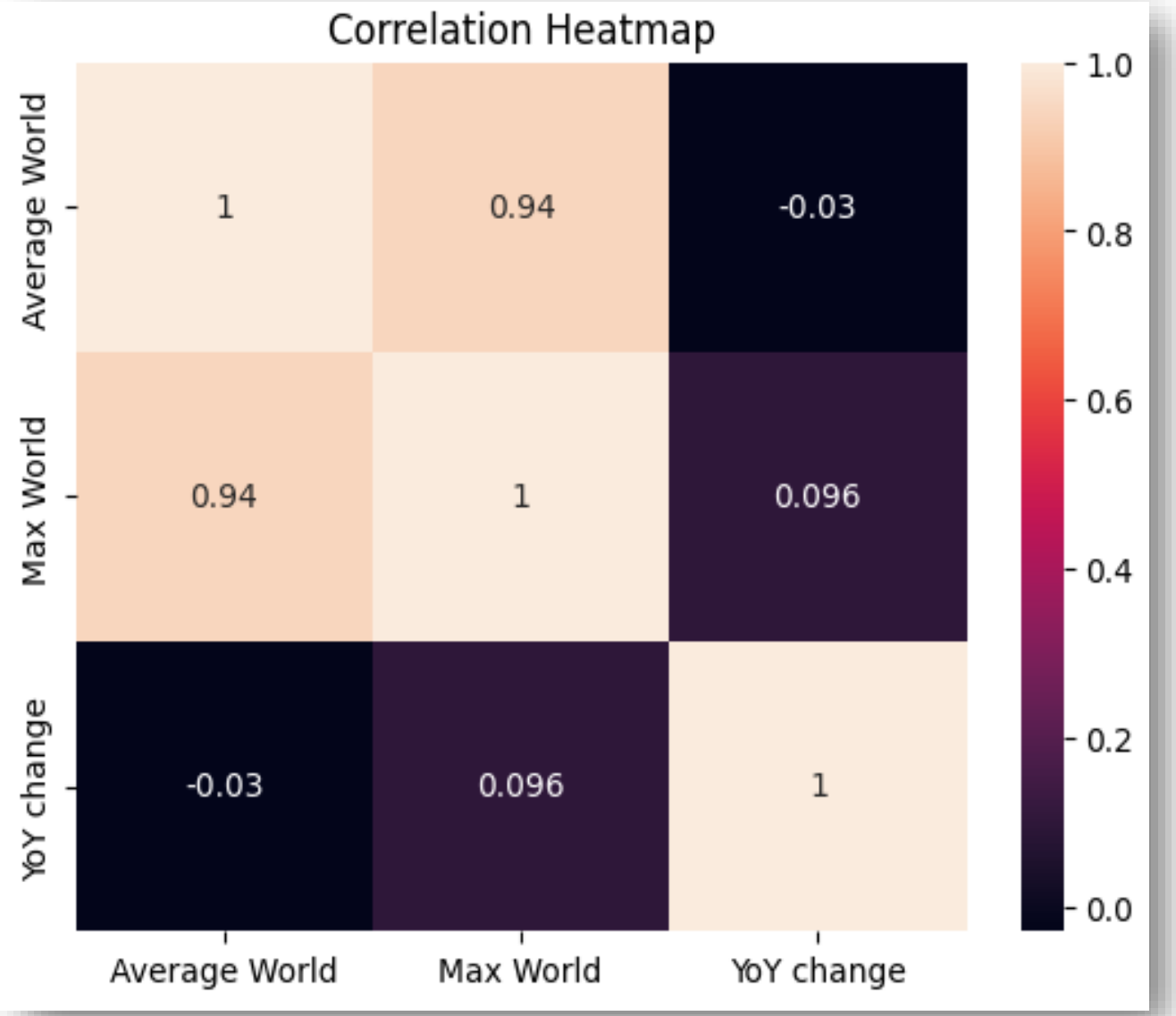
❑ YoY Change vs Average Unemployment

- No strong relationship is visible between YoY change and unemployment.
- Some outliers exist where YoY change is very high but unemployment is low.
- Overall, YoY change does not significantly affect average unemployment



❑ Correlation Heatmap

- Average World and Max World unemployment have a very strong positive correlation (0.94).
- YoY Change and Average World unemployment show almost no correlation (-0.03).
- YoY Change and Max World unemployment show very weak correlation (0.096).



CONCLUSION

This project analyzed global unemployment trends from 1998 to 2022 using Python data visualization. The graphs helped compare unemployment rates across countries, identify trends over time, and understand relationships between different unemployment indicators.

Code link:https://colab.research.google.com/drive/1fgLZ-r4SaNB-vEvyWS30q6kuNolE3ea3?usp=sharing

THANK YOU